# Training Dynamics of Overparameterized Neural Networks: A Study of Spectral Bias on $S^1$

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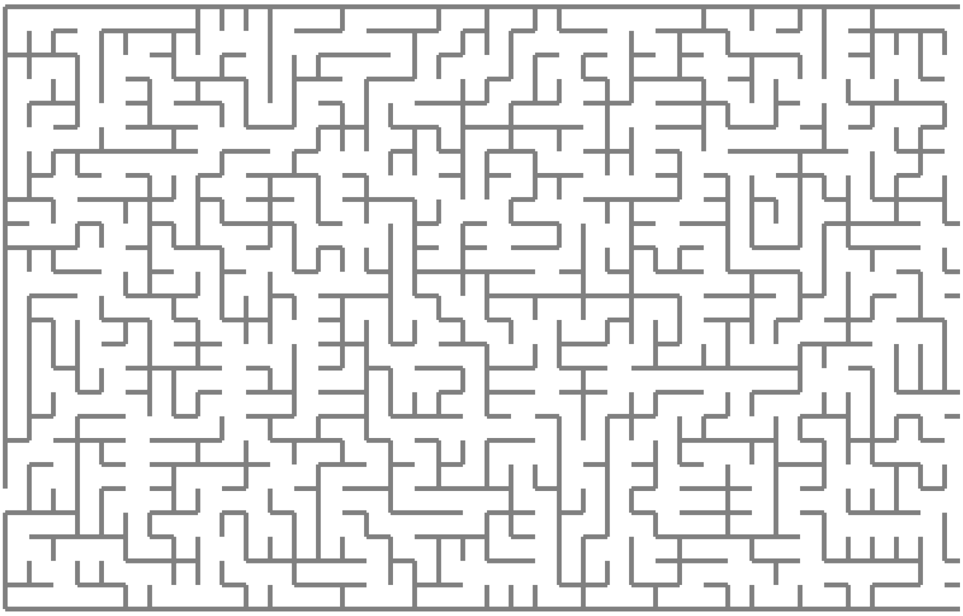

Shreyas Kalvankar

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THESIS

submitted in partial fulfillment of the
requirements for the degree of

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in

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by

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Cover picture: A "random" maze generated in postscript.

# Training Dynamics of Overparameterized Neural Networks: A Study of Spectral Bias on $S^1$

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**Abstract**

This thesis studies the training dynamics of overparameterized neural networks through spectral bias, the tendency of gradient-based training to learn low-frequency components of a target function faster than high-frequency ones. The setting is regression on the unit circle using a one-hidden-layer ReLU network, with targets given by finite Fourier mixtures.

The analysis begins in the infinite-width regime. In this limit, the operator governing the residual dynamics is the neural tangent kernel. Under the uniform measure on the circle, this operator is a convolution and is therefore diagonalized by the Fourier basis. This gives an explicit description of the residual dynamics and of the corresponding frequency-dependent learning rates.

The main contribution of the current thesis is a finite-sample analysis of this continuum operator. On a fixed low-frequency Fourier subspace, non-asymptotic bounds are derived that show the sampled operator remains close to the continuum operator with high probability. The final part of the project, still in progress, studies the effect of finite width in the continuum setting.

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# Preface

This is where you thank people for helping you etc.

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April 14, 2026

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# Chapter 1

# Introduction

Deep neural networks have achieved remarkable empirical success across a wide range of application domains, most notably in computer vision and natural language processing. In vision, deep convolutional and residual architectures have led to dramatic improvements on large-scale benchmarks, establishing deep learning as the dominant paradigm for visual recognition tasks (Krizhevsky et al., 2012; He et al., 2016). Similarly, in natural language processing, attention-based architectures introduces by Vaswani et al. (2023) such as Transformers have enabled substantial advances in sequence modeling and representation learning, and now form the foundation of modern large-scale language models. Over the past decade, this empirical progress has been accompanied by a steady increase in model size, depth, and computational scale, often resulting in highly overparameterized models that generalize well despite their capacity to fit random labels (Zhang et al., 2017; Belkin et al., 2019). Much of this progress has been driven by empirical experimentation and architectural intuition rather than by a complete theoretical understanding of the mechanisms underlying training and generalization in deep neural networks.

Despite their practical success, the mechanisms governing the training dynamics and convergence behavior of neural networks remain only partially understood, even in highly simplified settings. From an optimization perspective, neural network training involves high-dimensional, non-convex objectives whose geometry depends intricately on architectural choices, initialization, and optimization dynamics. Early analyses of neural network loss landscapes have highlighted the prevalence of saddle points and complex critical structures, underscoring the difficulty of directly characterizing training dynamics in parameter space Choromanska et al. (2015). While more recent theoretical work has made progress in analyzing overparameterized models under restrictive assumptions such as two-layer networks or specific scaling regimes these results do not yet provide a unified explanation of neural network training in general settings Arora et al. (2019). More broadly, this gap between empirical performance and theoretical understanding has motivated the study of neural networks through simplified models and asymptotic limits, where training dynamics can be analyzed more precisely.

## 1.1 Problem setup

**The Statistical learning framework** The fundamental objective in supervised learning is to identify a functional relationship between an input space $\mathcal{X} \subseteq \mathbb{R}^d$ and a target space $\mathcal{Y}$. We assume the existence of a joint probability distribution $\rho$ over $\mathcal{X} \times \mathcal{Y}$. Ideally, one seeks a measurable function $f : \mathcal{X} \to \mathcal{Y}$, that minimizes the expected risk (or population risk):

$$\mathcal{L}_\rho(f) = \mathbb{E}_\rho \ell(f(x), y) \tag{1.1}$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty]$ is a loss function quantifying the discrepancy between the prediction and the true target. In practice, two fundamental constraints prevent the direct minimization of the equation (1.1). First, searching over the space of all possible measurable functions is computationally and analytically intractable, so we usually restrict our search to a specific hypothesis class which is a family a functions $f_\theta$ parameterized by a vector $\theta \in \mathbb{R}^p$. Second, the underlying distribution $\rho$ is unknown; we only have access to a finite training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ of $n$ samples drawn i.i.d from $\rho$.

**Empirical risk minimization.** The second constraint leads us to the principle of empirical risk minimization (ERM). Given a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, a measurable function $f_\theta$ parameterized by $\theta$, and a loss function $\ell(\cdot, \cdot)$, ERM seeks parameters that minimize the empirical risk,

$$\theta^\star \in \arg\min_\theta \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i)\,. \tag{1.2}$$

This formulation isolates the key ingredients that determine the behavior of learning algorithms: the function class induced by the parameterization $\theta \mapsto f_\theta$, the geometry of the loss landscape, and the optimization procedure used to minimize the empirical risk. In modern neural network training, the empirical risk is typically minimized using gradient-based methods, which induce a dynamical system in parameter space whose properties depend intricately on both the model architecture and the chosen loss function.

**Linear models and gradient descent.** To build intuition, we first consider the case of linear models. We define a function $f_w(x) : \mathbb{R}^d \to \mathbb{R}$, parameterized by $w \in \mathbb{R}^d$ with the prescription $f_w(x) = x^\top w$. We specialize the loss function to the squared error, $\ell(f(x), y) = \frac{1}{2}(f_w(x) - y)^2$. Let $X \in \mathbb{R}^{d \times n}$ denote the data matrix whose columns are $x_i$, and let $y \in \mathbb{R}^n$ denote the vector of targets. With slight abuse of notation, the ERM problem simplifies to a least-squares objective:

$$\arg\min_w \mathcal{L}(w) = \frac{1}{2} \sum_{i=1}^n (f_w(x_i) - y_i)^2 = \frac{1}{2} \|X^\top w - y\|_2^2. \tag{1.3}$$

This objective is convex and differentiable, with gradient

$$\nabla_w \mathcal{L}(w) = X(X^\top w - y) = XX^\top w - Xy. \tag{1.4}$$

At a stationary point $w^\star$, the gradient vanishes, yielding the equations

$$XX^\top w^\star = Xy, \tag{1.5}$$

which characterize the set of global minimizers of $\mathcal{L}$.
The structure of the minimizers depends on the rank of the empirical covariance operator $A := XX^\top$. If $A$ is positive definite (equivalently, if $X$ has full row rank), then the minimizer is unique and given by $w^\star = A^{-1}Xy$. In contrast, in the overparameterized regime where $\mathrm{rank}(X) < d$, the matrix $A$ is singular and the least-squares objective admits infinitely many global minimizers. In this case, the Moore–Penrose pseudoinverse can be used to approximate a solution. Since $A$ is symmetric, its pseudoinverse $A^+$ acts as the inverse on $\mathrm{range}(A)$ and annihilates $\ker(A)$, yielding the minimum-norm least-squares solution $w^\star = A^+Xy$.
Gradient descent with step size $\eta > 0$ takes the explicit form

$$w_{t+1} = w_t - \eta\nabla_w\mathcal{L}(w_t) = \left(I - \eta XX^\top\right)w_t + \eta Xy. \tag{1.6}$$

Assuming $w_0 = 0$, we can write a closed form solution

$$w_t = \eta\sum_{s=0}^{t-1}\left(I - \eta XX^\top\right)^s Xy. \tag{1.7}$$

Let $\lambda_{\max}(A)$ denote the largest eigenvalue of $A$. If the step size satisfies $\eta \in (0, 2/\lambda_{\max}(A))$, then for any least-squares minimizer $w^\star$ the error $e_t := w_t - w^\star$ evolves as

$$e_{t+1} = (I - \eta A)e_t. \tag{1.8}$$

Since $A$ is symmetric, it admits an eigendecomposition

$$A = Q\Lambda Q^\top,$$

where $Q \in \mathbb{R}^{d\times d}$ is orthogonal ($Q^\top Q = I$) and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ contains the (real) eigenvalues of $A$. Moreover, $A$ is positive semidefinite because $v^\top Av = \|X^\top v\|_2^2 \geq 0$ for all $v$, hence $\lambda_i \geq 0$ and in particular $\lambda_i \in [0, \lambda_{\max}(A)]$.

Starting from the error recursion

$$e_{t+1} = (I - \eta A)e_t,$$

we express the error in the eigenbasis of $A$ by defining $\tilde{e}_t := Q^\top e_t$. Left-multiplying by $Q^\top$ and using $e_t = Q\tilde{e}_t$ gives

$$\tilde{e}_{t+1} = Q^\top(I - \eta A)Q\tilde{e}_t = \left(I - \eta Q^\top AQ\right)\tilde{e}_t = \left(I - \eta\Lambda\right)\tilde{e}_t.$$

Since $\Lambda$ is diagonal, this yields the component-wise dynamics

$$\tilde{e}_{t+1,i} = (1 - \eta\lambda_i)\tilde{e}_{t,i}, \qquad i = 1, \ldots, d. \tag{1.9}$$

If $\eta \in (0, 2/\lambda_{\max}(A))$, then for any $\lambda_i > 0$ we have $0 < \eta\lambda_i \leq \eta\lambda_{\max}(A) < 2$, hence $-1 < 1 - \eta\lambda_i < 1$, i.e. $|1 - \eta\lambda_i| < 1$. Under the stated condition on $\eta$, we have $|1 - \eta\lambda_i| < 1$ for all $\lambda_i > 0$, implying geometric decay of all components of $e_t$ in $\mathrm{range}(A)$. In particular, if $A$ is positive definite, then $w_t \to w^\star$ as $t \to \infty$. More generally, if $w_0 \in \mathrm{range}(A)$ (for example, $w_0 = 0$), then $w_t$ converges to the minimum-norm least-squares solution $w^\star = A^+Xy$.

The eigen-decomposition above shows that gradient descent acts as a linear dynamical system whose behavior is governed by the spectrum of the empirical covariance operator $XX^\top$. Directions corresponding to larger eigenvalues converge more rapidly, while components in the null space of $XX^\top$ remain unchanged. In the overparameterized regime, where multiple *interpolating solutions* exist, i.e., solutions satisfying $X^\top w = y$, gradient descent from standard initializations converges to a particular interpolating solution, namely the minimum-norm solution. This illustrates how the optimization algorithm induces an implicit bias even in this simplest linear setting.

**Deep linear network.** Even in the absence of nonlinear activation functions, introducing an additional layer already leads to substantially more intricate training dynamics. Consider a two-layer linear network with scalar output,

$$f_{v,W}(x) := v^\top W x, \tag{1.10}$$

where $W \in \mathbb{R}^{m\times d}$, $v \in \mathbb{R}^m$, and $m$ denotes the width of the hidden layer. The least-squares empirical risk is

$$\mathcal{L}(v,W) := \frac{1}{2}\left\|X^\top W^\top v - y\right\|_2^2. \tag{1.11}$$

Although the resulting function is linear in the input, the loss is no longer convex in the parameters $(v,W)$ due to their multiplicative coupling.

Let

$$r := X^\top W^\top v - y \in \mathbb{R}^n \tag{1.12}$$

denote the residual vector. The gradients of $\mathcal{L}$ are given by

$$\nabla_v \mathcal{L} = WX\,r, \qquad \nabla_W \mathcal{L} = v(Xr)^\top. \tag{1.13}$$

Similar to the linear case, the parameters of this model are updated iteratively using gradient descent with step size $\eta > 0$

$$v_{k+1} = v_k - \eta\nabla_v \mathcal{L}(v_k,W_k), \qquad W_{k+1} = W_k - \eta\nabla_W \mathcal{L}(v_k,W_k).$$

The analysis of learning dynamics is often simplified by considering the infinitesimal learning rate limit ($\eta \to 0$). In this regime, the discrete updates converge to a system of coupled ordinary differential equations (ODEs) known as gradient flow.

$$\dot{v}_t = -\nabla_v \mathcal{L}(v_t,W_t) = W_t X\left(y - X^\top W_t^\top v_t\right), \quad \dot{W}_t = -\nabla_W \mathcal{L}(v_t,W_t) = v_t\left(y - X^\top W_t^\top v_t\right)^\top X^\top. \tag{1.14}$$

This coupled system consists of a vector-valued and a matrix-valued ordinary differential equation and is nonlinear in the parameters, despite the underlying predictor being linear in the input. As a result, the training dynamics no longer admit a simple spectral characterization as in the one-layer case.

Nevertheless, recent work has shown that deep linear networks admit a more structured description in suitable asymptotic regimes. In particular, Chizat et al. (2024) study gradient flow for deep linear networks in an infinite-width limit, where the dynamics are described at the level of a measure-valued evolution. While no closed-form solution of the finite-dimensional ODE system is obtained, this framework establishes global well-posedness, convergence to global minimizers, and the emergence of implicit regularization effects induced by depth.

**Nonlinear networks.** The analysis of training dynamics becomes more involved once nonlinear activation functions are introduced. Consider again a two-layer network of the form

$$f_{v,W}(x) = v^\top \phi(Wx),$$

where $\phi : \mathbb{R} \to \mathbb{R}$ is a continuously differentiable activation function applied elementwise. For the least-squares objective, the continuous-time evolution of the parameters is governed by the gradient flow system:

$$\dot{v}_t = \phi(W_t X)(y - \phi(W_t X)^\top v_t), \qquad \dot{W}_t = \Big(\phi(W_t X) \odot \Big(v_t (y - \phi(W_t X)^\top)^\top\Big)\Big) X^\top, \tag{1.15}$$

where $\odot$ denotes the Hadamard (elementwise) product. The gradient flow equations involve both the activations $\phi(Wx)$ and their derivatives $\phi'(Wx)$. In particular, the evolution of the parameters mixes standard matrix products with elementwise nonlinear operations. As a consequence, the training dynamics of nonlinear networks do not admit a simple closed-form or spectral description in general.

**Thesis motivation.** At a high level, this thesis is motivated by the question of how these networks' training dynamics can be studied in a mathematically tractable setting without discarding the phenomena one ultimately seeks to understand. The objective is not to provide a complete description of neural network training in full generality, but rather to identify a setting in which the dynamics can be analyzed explicitly while still retaining features that remain meaningful for finite networks and finite datasets.

This motivates the study of analytical regimes in which the dynamics become tractable. Although such regimes necessarily involve simplifications like taking large-width limits, they can still provide useful insight into mechanisms that are otherwise difficult to access in the full non-linear parameter dynamics. Two prominent examples of such approaches are the neural tangent kernel and mean-field formulations, which offer complementary perspectives on the behavior of wide neural networks during training.

# Chapter 2

# Related Work

## 2.1 Neural Tangent Kernel

As discussed in Chapter 1.1, for nonlinear neural networks the gradient flow dynamics are parameter-dependent and do not admit a simple closed-form description. One approach to recovering a tractable analytical framework is to consider regimes in which the network behaves approximately linearly around its random initialization. The *Neural Tangent Kernel* (NTK), introduced by Jacot et al. (2020), formalizes this idea by studying training dynamics through a first-order linearization in parameter space, leading to a kernel-based description of learning in wide neural networks.

### 2.1.1 Linearization around initialization

Let $f(x;\theta)$ denote a neural network with parameters $\theta \in \mathbb{R}^p$, initialized at $\theta_0$. A first-order Taylor expansion around $\theta_0$ yields

$$f(x;\theta) \;\approx\; f(x;\theta_0) + \nabla_\theta f(x;\theta_0)^\top (\theta - \theta_0), \tag{2.1}$$

where $f(x;\theta_0)$ is the network output at initialization and

$$\phi(x) := \nabla_\theta f(x;\theta_0)$$

defines the tangent feature map. Locally around initialization, the network thus behaves as a linear model in parameter space,

$$f(x;\theta) \approx f(x;\theta_0) + \phi(x)^\top (\theta - \theta_0). \tag{2.2}$$

**Definition 2.1** (Neural Tangent Kernel (Jacot et al., 2020))**.** Given an initialization $\theta_0$, the neural tangent kernel is defined as

$$\Theta_0(x,x') := \nabla_\theta f(x;\theta_0)^\top \nabla_\theta f(x';\theta_0).$$

The NTK measures how infinitesimal parameter updates couple the network outputs at different inputs and plays the role of an empirical covariance operator in the tangent feature space.

### 2.1.2 Training dynamics induced by the NTK

We consider training with the squared loss using gradient descent with step size $\eta > 0$,

$$L(\theta) = \frac{1}{2}\sum_{i=1}^{n}\big(f(x_i;\theta) - y_i\big)^2, \qquad \theta_{t+1} = \theta_t - \eta\nabla_\theta L(\theta_t), \qquad t = 0,1,2,\ldots$$

Define $f_t(x) := f(x;\theta_t)$. The gradient of the loss is

$$\nabla_\theta L(\theta_t) = \sum_{j=1}^{n}\big(f_t(x_j) - y_j\big)\,\nabla_\theta f(x_j;\theta_t).$$

Applying a first-order Taylor expansion of $f(\cdot;\theta)$ around $\theta_t$ gives

$$f_{t+1}(x_i) \approx f_t(x_i) + \nabla_\theta f(x_i;\theta_t)^\top(\theta_{t+1} - \theta_t).$$

Substituting the gradient descent update $\theta_{t+1} - \theta_t = -\eta\nabla_\theta L(\theta_t)$ yields

$$f_{t+1}(x_i) = f_t(x_i) - \eta\sum_{j=1}^{n}\nabla_\theta f(x_i;\theta_t)^\top\nabla_\theta f(x_j;\theta_t)\,\big(f_t(x_j) - y_j\big),$$

which can be written compactly as

$$f_{t+1}(x_i) = f_t(x_i) - \eta\sum_{j=1}^{n}\Theta_t(x_i,x_j)\big(f_t(x_j) - y_j\big), \tag{2.3}$$

where the iteration-dependent NTK is

$$\Theta_t(x_i,x_j) := \nabla_\theta f(x_i;\theta_t)^\top\nabla_\theta f(x_j;\theta_t).$$

If we stack predictions into $f_t := (f_t(x_1),\ldots,f_t(x_n))^T \in \mathbb{R}^n$, in vector form, the dynamics read

$$f_{t+1} = f_t - \eta\,\Theta_t(f_t - y). \tag{2.4}$$

where $\bar{\Theta} = \big(\bar{\Theta}(x_i,x_j)\big)_{i,j=1}^{n} \in \mathbb{R}^{n\times n}$ is the kernel gram matrix. In general, $\Theta_t$ evolves during training, reflecting changes in the network's tangent features.

### 2.1.3 One-hidden-layer NTK and infinite-width limit

To make the NTK explicit, consider a two-layer network of width $m$,

$$f(x) = \frac{1}{\sqrt{m}}\sum_{\alpha=1}^{m} v_\alpha\,\varphi(w_\alpha^\top x), \qquad v_\alpha \in \mathbb{R}, \quad w_\alpha, x \in \mathbb{R}^d. \tag{2.5}$$

Each neuron is parameterized by $\theta_\alpha = (v_\alpha, w_\alpha)$ and initialized as

$$v_\alpha \sim \mathcal{N}(0,\sigma_v^2), \qquad w_\alpha \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{d}I_d\right),$$

independently across $\alpha = 1,\ldots,m$. $\varphi : \mathbb{R} \to \mathbb{R}$ is a non-linear activation function.

**Finite-width NTK at initialization.** By direct computation,

$$\nabla_{v_\alpha} f(x) = \frac{1}{\sqrt{m}} \varphi(w_\alpha^\top x), \tag{2.6}$$

$$\nabla_{w_\alpha} f(x) = \frac{1}{\sqrt{m}} v_\alpha \, \varphi'(w_\alpha^\top x) \, x. \tag{2.7}$$

Substituting into the definition of the NTK yields the empirical kernel

$$\Theta_0^{(m)}(x,x') = \frac{1}{m} \sum_{\alpha=1}^{m} \varphi(w_\alpha^\top x)\, \varphi(w_\alpha^\top x') + \frac{1}{m} \sum_{\alpha=1}^{m} v_\alpha^2 \, \varphi'(w_\alpha^\top x)\, \varphi'(w_\alpha^\top x')\, x^\top x'. \tag{2.8}$$

**Infinite-width limit.** Each sum in (2.8) is an empirical average of i.i.d. terms. By the strong law of large numbers,

$$\Theta_0^{(m)}(x,x') \xrightarrow{\text{a.s.}} \bar{\Theta}(x,x') \qquad \text{as } m \to \infty,$$

where the limiting NTK is deterministic and given by

$$\bar{\Theta}(x,x') = \mathbb{E}_w\big[\varphi(w^\top x)\, \varphi(w^\top x')\big] + \sigma_v^2 \, x^\top x' \, \mathbb{E}_w\big[\varphi'(w^\top x)\, \varphi'(w^\top x')\big]. \tag{2.9}$$

This argument extends to deep fully connected networks, where both the forward covariance kernel and the NTK satisfy recursive layerwise equations in the infinite-width limit (Jacot et al., 2020; Lee et al., 2020).

### 2.1.4 Constant-kernel regime and spectral dynamics

In the infinite-width limit, the NTK remains constant throughout training, $\Theta_t \equiv \bar{\Theta}$. Equation (2.4) then reduces to the linear iteration

$$f_{t+1} = f_t - \eta \, \bar{\Theta}(f_t - y). \tag{2.10}$$

Letting $r_t := f_t - y$, we obtain

$$r_{t+1} = (I - \eta\bar{\Theta})\, r_t.$$

Assuming $\eta < 2/\lambda_{\max}(\bar{\Theta})$, the residual admits the closed form

$$r_t = (I - \eta\bar{\Theta})^t r_0, \qquad f_t = y + (I - \eta\bar{\Theta})^t (f_0 - y). \tag{2.11}$$

As in the linear setting of Chapter 1.1, convergence occurs independently along the eigenvectors of $\bar{\Theta}$, with geometric rates determined by the corresponding eigenvalues.

### 2.1.5 NTK dynamics, solution structure, and implicit bias

In the constant-kernel (infinite-width) regime, the NTK prediction dynamics reduce to a linear iteration in prediction space. Eq 2.10, admits fixed points $f^\star$ satisfying

$$\bar{\Theta}\,(f^\star - y) = 0. \tag{2.12}$$

Equation (2.12) characterizes the set of global minimizers of the squared loss in the NTK regime. As in linear least squares, the structure of this solution set depends on the rank of the operator $\bar{\Theta}$.

If $\bar{\Theta}$ is strictly positive definite on the training set, the solution is unique and satisfies $f^\star = y$. When $\bar{\Theta}$ is singular, the loss admits infinitely many interpolating solutions in prediction space, differing by elements of $\ker(\bar{\Theta})$. In this case, gradient descent converges to a particular fixed point determined by the initialization.

**Tangent features and operator factorization.** To formulate the NTK in a way that remains meaningful in the infinite-width limit, it is convenient to view parameter perturbations as elements of a Hilbert space $\mathcal{H}_\theta$ equipped with an inner product $\langle \cdot, \cdot \rangle$. In the finite-dimensional case, $\mathcal{H}_\theta = \mathbb{R}^p$ with the Euclidean inner product; in the infinite-width limit, $\mathcal{H}_\theta$ denotes the corresponding limit space of parameter perturbations, often referred to as the tangent parameter space (Jacot et al., 2020; Lee et al., 2020).

In this setting, the limiting NTK Gram matrix on the training set can be expressed as

$$\bar{\Theta} = J_0 J_0^*, \tag{2.13}$$

where $J_0 : \mathcal{H}_\theta \to \mathbb{R}^n$ denotes the Jacobian of the network outputs with respect to parameters at initialization, viewed as a linear operator,

$$(J_0 h)_i = \langle \nabla_\theta f(x_i; \theta_0), h \rangle,$$

and $J_0^*$ is its adjoint. This representation makes explicit that $\bar{\Theta}$ is symmetric and positive semidefinite, and that it acts as an empirical covariance operator for the tangent features induced by the network at initialization (Jacot et al., 2020).

**Implicit bias and representer viewpoint.** When $\bar{\Theta}$ is singular, the Moore–Penrose pseudoinverse $\bar{\Theta}^+$ acts as the inverse on $\mathrm{range}(\bar{\Theta})$ and annihilates $\ker(\bar{\Theta})$. Consequently, constant-kernel NTK training eliminates only the residual component lying in $\mathrm{range}(\bar{\Theta})$ while preserving the component in $\ker(\bar{\Theta})$. This behavior reflects an implicit regularization effect analogous to linear least squares and can be interpreted as convergence to a minimum-norm solution in the reproducing kernel space associated with $\bar{\Theta}$, as formalized by representer-theorem results (Bietti and Mairal, 2019; Bartolucci et al., 2021).

**Limitations of stable kernels and feature learning.** Although the NTK framework provides a clean and tractable description of training dynamics, the stability of the kernel also introduces inherent limitations. When the kernel does not change during training, learning is effectively confined to a fixed feature space. Indeed, any positive semidefinite kernel admits a representation $K(x,x') = \langle \Phi(x), \Phi(x') \rangle_{\mathcal{H}}$, so that optimization in the NTK regime corresponds to fitting a linear model in the associated reproducing kernel Hilbert space $\mathcal{H}$.

In contrast, for finite-width networks the NTK typically evolves during training, implying that the tangent features $\Phi_t(x) = \nabla_\theta f(x; \theta_t)$ also change over time. This evolution allows the network to adapt its representation to the data, a phenomenon commonly referred to as *feature learning*. Such effects are absent in a strictly stable-kernel regime, where only the coefficients of fixed features are adjusted.

The importance of feature learning is already visible in simplified settings. Even purely linear networks exhibit nontrivial training dynamics due to the coupling of parameters across layers, leading to implicit biases that cannot be captured by a fixed kernel model (Saxe et al., 2014; Tu et al., 2024). From this viewpoint, the NTK regime can be seen as a useful but restrictive approximation, which motivates studying regimes where kernel evolution and feature learning play an explicit role.

## 2.2 Mean-field representation of a two-layer network

While the neural tangent kernel provides a linearized description of training dynamics around initialization, it does not capture regimes in which features evolve significantly during optimization. An alternative and complementary approach is provided by the *mean-field* perspective, which studies wide neural networks by viewing their parameters as interacting particles and describing training as the evolution of a probability distribution over parameter space.

As the network width tends to infinity under appropriate scalings, the training dynamics induced by gradient descent converge to a deterministic evolution equation for the parameter distribution. This viewpoint was first formalized for two-layer networks by Mei et al. (2018), who derived a nonlinear partial differential equation governing the evolution of the parameter measure. Related interacting particle system formulations and convergence results were developed by Rotskoff and Vanden-Eijnden (2022), while extensions to multilayer networks were proposed by Nguyen (2019). From an optimization perspective, Chizat and Bach (2018) showed that, under suitable initialization and in the many-particle limit, training dynamics can be interpreted as a Wasserstein gradient flow on the space of probability measures. Informally, this means that instead of tracking individual neurons, one tracks how their overall distribution moves over parameter space during training. The evolution is governed by a transport equation, so learning is viewed as a continuous redistribution of mass that decreases the loss. In this formulation, the evolution of the network is described directly at the level of measures, and the flow converges to global minimizers despite the non-convexity of the finite-dimensional parameterization.

A detailed measure-theoretic derivation of the mean-field representation is out of the scope of this thesis and is deferred to Appendix A.1.4. Here we only record the aspects that are relevant for contrasting the mean-field viewpoint with the NTK regime.

Consider the two-layer neural network from (2.5) with modified scaling of $\frac{1}{m}$

$$f_\theta(x) = \frac{1}{m}\sum_{\alpha=1}^{m} v_\alpha \varphi(w_\alpha^\top x), \qquad v_\alpha \in \mathbb{R},\ w_\alpha, x \in \mathbb{R}^d, \tag{2.14}$$

where each neuron is parameterized by $\theta_\alpha = (v_\alpha, w_\alpha)$, $\alpha \in \{1, \ldots, m\}$.

### 2.2.1 Integral representation and empirical measure

Let $\Omega := \mathbb{R} \times \mathbb{R}^d$ denote the parameter space, and define the empirical neuronal measure

$$\mu_m := \frac{1}{m} \sum_{\alpha=1}^{m} \delta_{(v_\alpha, w_\alpha)}. \tag{2.15}$$

Then $\mu_m \in \mathcal{P}(\Omega)$, where $\mathcal{P}(\Omega)$ is the space of Borel probability measures on $\Omega$. A detailed derivation is given in Appendix A.1.1.

With this notation, the network output can be written as

$$f_\theta(x) = \int_\Omega v \varphi(w^\top x)\, d\mu_m(v, w). \tag{2.16}$$

Thus, under the mean-field parameterization, the network depends on its parameters only through their empirical distribution.

**Mean-field viewpoint.** Equation (2.16) shows that the network output can be written as a linear functional of the empirical measure $\mu_m$. This representation is only possible because we scale the output by $1/m$ and because the hidden neurons are permutation invariant. Such observations motivate the mean-field viewpoint, in which the empirical measure $\mu_m$ is regarded as the fundamental state variable describing the network (Mei et al., 2018; Chizat and Bach, 2018).

**Initialization and limiting representation.** Assume that the parameters $\{\theta_\alpha(0)\}_{\alpha=1}^m$ are initialized i.i.d. according to a probability measure $\mu_0$ on $\Omega$. Then the empirical measure at initialization converges weakly to $\mu_0$ almost surely as $m \to \infty$. Consequently, the network output at initialization converges pointwise to the deterministic limit

$$f_{\mu_0}(x) := \int_\Omega v \varphi(w^\top x)\, d\mu_0(v, w).$$

For instance, under i.i.d. Gaussian initialization of the parameters, the limiting initial measure $\mu_0$ is Gaussian on $\Omega$.

### 2.2.2 Training dynamics and evolving kernel

Let $X = (x_1, \ldots, x_n) \in \mathbb{R}^{d \times n}$ and $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$ denote the training set, and consider the squared loss

$$L(v, W) = \frac{1}{2} \sum_{i=1}^{n} \big(f_{v,W}(x_i) - y_i\big)^2.$$

Under gradient flow, the particle system induces an evolution of the empirical measure $\mu_{m,t}$ in parameter space. More precisely, there is an explicit measure-dependent velocity field

$$b((v, w); \mu) := \begin{pmatrix} \varphi(X^\top w)^\top \big(y - f_\mu(X)\big) \\ X\Big(\varphi'(X^\top w) \odot \big(y - f_\mu(X)\big)\Big) v \end{pmatrix},$$

where

$$f_\mu(x) := \int_\Omega v\,\varphi(w^\top x)\,d\mu(v,w), \qquad f_\mu(X) := \big(f_\mu(x_i)\big)_{i=1}^n,$$

such that the particle dynamics can be written as

$$\dot{\theta}_{t,\alpha} = \frac{\eta}{m}\,b(\theta_{t,\alpha};\mu_{m,t}), \qquad \alpha = 1,\ldots,m.$$

A direct derivation of this system is given in Appendix A.1.3.

Equivalently, the empirical measure solves the continuity equation

$$\partial_t \mu_{m,t} + \nabla_\theta \cdot \Big(\mu_{m,t}\,\frac{\eta}{m}\,b(\cdot;\mu_{m,t})\Big) = 0. \tag{2.17}$$

A weak formulation and the associated push-forward representation are given in Appendix A.1.4.

Choosing the mean-field scaling $\eta = m$ yields a nontrivial $O(1)$ evolution in time and, as $m \to \infty$, one expects $\mu_{m,t} \to \mu_t$, where the limit $\mu_t$ satisfies

$$\partial_t \mu_t + \nabla_\theta \cdot \big(\mu_t\,b(\cdot;\mu_t)\big) = 0. \tag{2.18}$$

**Advantages over the NTK limit.** Recall the finite-width NTK expression (2.8). In the mean-field parameterization $f(x) = \frac{1}{m}\sum_{\alpha=1}^m v_\alpha \varphi(w_\alpha^\top x)$, the empirical tangent kernel is

$$\Theta_t^{(m)}(x,x') = \frac{1}{m^2}\sum_{\alpha=1}^m \Big(\varphi(w_{t,\alpha}^\top x)\,\varphi(w_{t,\alpha}^\top x') + v_{t,\alpha}^2\,\varphi'(w_{t,\alpha}^\top x)\,\varphi'(w_{t,\alpha}^\top x')\,x^\top x'\Big). \tag{2.19}$$

Since $\Theta_t^{(m)} = O(m^{-1})$, the effective operator driving learning is the scaled kernel $\eta\,\Theta_t^{(m)}$. Under the mean-field scaling $\eta = m$, this quantity has a finite limit:

$$\lim_{m\to\infty}\big(\eta\,\Theta_t^{(m)}(x,x')\big) = \int_\Omega \Big(\varphi(w^\top x)\,\varphi(w^\top x') + v^2\,\varphi'(w^\top x)\,\varphi'(w^\top x')\,x^\top x'\Big)\,d\mu_t(v,w).$$

As $\mu_t$ evolves according to the mean-field transport equation, this limiting kernel also evolves in time. In contrast to the frozen-kernel NTK regime, the mean-field viewpoint therefore captures feature learning through an explicitly time-dependent limiting representation.

**Disadvantages.** The mean-field scaling yields a deterministic measure-valued evolution $(\mu_t)_{t\geq 0}$ as $m \to \infty$, but its long-time behavior is generally difficult to characterize. In particular, it is not clear in general whether $\mu_t$ converges as $t \to \infty$, nor which minimizer is selected at the level of measures. Moreover, unlike the NTK limit, extending the above measure-evolution construction beyond two-layer networks is not straightforward, since deeper architectures do not admit the same simple permutation-invariant description in terms of an empirical measure on a fixed parameter space. Several multilayer mean-field-type constructions have been proposed to address this difficulty (Nguyen, 2019; Araújo et al., 2019; Sirignano and Spiliopoulos, 2019; Nguyen and Pham, 2023).

## 2.3 Spectral bias

A striking and widely reported phenomenon in neural network training is that gradient-based optimization does not fit all components of a target function at the same rate. Instead, networks tend to learn "simple" structure first and refine fine-scale or highly oscillatory structure later. This preference is commonly referred to as *spectral bias* (or the *frequency principle*). In its classical formulation, spectral bias asserts that, when a target function is decomposed into Fourier modes, lower-frequency components are learned earlier and at a faster rate than higher-frequency components (Rahaman et al., 2019).

In this section we briefly review empirical evidence for spectral bias and summarize theoretical viewpoints that relate it to the spectrum of the operator governing training dynamics in the kernel (NTK) regime. We also highlight extensions that study how the phenomenon depends on the input distribution and how it manifests outside the training set.

### 2.3.1 Empirical evidence and experimental protocols

A standard way to empirically probe spectral bias is to choose a target with a controlled frequency decomposition and to track the frequency content of the network's prediction $f_t$ during training.

**Synthetic Fourier regression.** A canonical experiment is 1D regression on a target constructed as a sum of sinusoids,

$$y(z) = \sum_{i=1}^{r} A_i \sin(2\pi k_i z + \phi_i), \qquad z \in [0,1],$$

and monitoring the discrete Fourier spectrum of the learned predictor as a function of training time. Rahaman et al. (2019) report that, for deep ReLU networks trained by (full-batch) gradient descent, Fourier coefficients at smaller frequencies $k_i$ grow substantially earlier than those at larger $k_i$, even when the amplitudes are matched or when high-frequency components have larger amplitude (Rahaman et al., 2019). This "frequency-dependent learning speed" is one of the most direct empirical signatures of spectral bias.

**Robustness and perturbation tests.** A complementary protocol is to train to near-zero training error and then apply random perturbations in parameter space, comparing how the Fourier spectrum of the realized function changes. Rahaman et al. (2019) observe that lower-frequency components of the learned function are substantially more robust to such perturbations than higher-frequency components, suggesting that the network parameterization itself represents low frequencies in a more stable manner (Rahaman et al., 2019).

**Distributional effects and "local" frequency learning.** Spectral bias is often presented under uniformly sampled inputs, but empirical studies also examine how it interacts with the input distribution. For example, Basri et al. (2020) compare learning under uniform and non-uniform sampling densities and show that the ordering "low frequencies before high frequencies" can be modulated by where samples concentrate: dense regions of the input space may exhibit faster learning of higher-frequency structure than sparse regions (Basri et al., 2020). This line of experimentation motivates viewing spectral bias as a property of an operator defined jointly by the model and the data distribution, rather than as a purely architectural effect.

### 2.3.2 Operator spectrum viewpoint and the NTK regime

A common theoretical explanation of spectral bias proceeds by identifying an operator whose eigendecomposition controls learning rates. In the kernel/NTK regime, training dynamics are approximately linear in function space and decompose along eigendirections of the corresponding kernel operator. In this setting, spectral bias can be interpreted more generally as a bias toward learning the leading eigenfunctions of the kernel (rather than specifically Fourier modes) (Bowman and Montufar, 2022).

**Decomposition along NTK eigenfunctions.** Cao et al. (2020) give a rigorous account of this mechanism in the NTK regime: the training process can be decomposed along eigenfunctions of the NTK operator, with each component converging at a rate determined by the associated eigenvalue. As a consequence, components aligned with larger eigenvalues are learned faster, yielding a principled notion of "spectral preference" (Cao et al., 2020).

In general, the eigenfunctions of the NTK operator depend jointly on the input distribution and the kernel, so they need not coincide with the standard Fourier basis. A simplification occurs in the *infinite-width* (kernel) limit, where the empirical NTK concentrates around a deterministic limit $\bar{\Theta}$ given by an expectation over the random initialization.

To illustrate how Fourier modes arise, consider inputs on the circle $S^1$, parameterized by $\theta \in [0, 2\pi)$ and embedded as $x(\theta) = (\cos\theta, \sin\theta) \in \mathbb{R}^2$. Assume isotropic initialization (e.g. Gaussian), so that $R^\top w$ has the same distribution as $w$ for every orthogonal matrix $R$. Writing $\bar{\Theta}$ as an expectation (cf. Eq. (2.9)) and using this rotational invariance in distribution implies

$$\bar{\Theta}(Rx, Rx') = \bar{\Theta}(x, x') \qquad \text{for all orthogonal } R.$$

Since $x(\theta+\alpha) = R_\alpha x(\theta)$ and $x(\theta)^\top x(\theta') = \cos(\theta-\theta')$, it follows that $\bar{\Theta}(\theta, \theta')$ depends only on $\theta - \theta'$; equivalently, there exists $\kappa$ such that

$$\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta').$$

**Convolution form on $S^1$.** Let $\rho$ be the uniform probability measure on $S^1$, $d\rho(\theta') = d\theta'/(2\pi)$, and define the integral operator

$$(T_{\bar{\Theta}} g)(\theta) := \int_0^{2\pi} \bar{\Theta}(\theta, \theta')\, g(\theta')\, \frac{d\theta'}{2\pi}.$$

Using $\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta')$ and the change of variables $u = \theta - \theta'$, we obtain

$$(T_{\bar{\Theta}} g)(\theta) = \int_0^{2\pi} \kappa(u)\, g(\theta - u)\, \frac{du}{2\pi} =: (\kappa * g)(\theta),$$

which is the (circular) convolution operator on $S^1$.

**Fourier eigenfunctions and frequency ordering.** Because $T_{\bar{\Theta}}$ is a convolution, the Fourier modes $\phi_q(\theta) = e^{iq\theta}$, $q \in \mathbb{Z}$, are the eigenfunctions:

$$(T_{\bar{\Theta}}\phi_q)(\theta) = e^{iq\theta} \int_0^{2\pi} \kappa(u) e^{-iqu} \frac{du}{2\pi} = \lambda_q \phi_q(\theta), \qquad \lambda_q = \int_0^{2\pi} \kappa(u) e^{-iqu} \frac{du}{2\pi}.$$

If $\kappa$ is real and even (as in $\kappa(u) = \psi(\cos u)$), then $\lambda_q \in \mathbb{R}$ and $\lambda_q = \lambda_{-q}$, and one may equivalently use the real basis $\{\cos(q\theta), \sin(q\theta)\}$. In kernel gradient dynamics, each mode contracts independently at a rate set by $\lambda_q$ (e.g. $(1 - \eta\lambda_q)^t$ in discrete time), so modes with larger eigenvalues are learned faster. In symmetric NTK settings on the circle/sphere, the eigenvalues decrease with frequency, leading to the characteristic "low frequencies first" behavior (Basri et al., 2020; Rahaman et al., 2019).

### 2.3.3 Uniform vs. non-uniform sampling

The Fourier-mode picture above relies not only on a shift-invariant kernel but also on the *uniform* input distribution on $S^1$. Under uniform sampling, $d\rho(\theta') = d\theta'/(2\pi)$, shift invariance $\bar{\Theta}(\theta, \theta') = \kappa(\theta - \theta')$ implies that the integral operator is a circular convolution and therefore diagonalizes in the Fourier basis. In this setting, spectral bias can be stated directly in terms of frequency: learning rates are controlled by the eigenvalues $\lambda_q$ associated with Fourier modes.

Under *non-uniform* sampling, this simplification breaks. If inputs are distributed according to a density $p(\theta)$ on $S^1$, then the relevant population operator becomes

$$(T_p g)(\theta) := \int_0^{2\pi} \kappa(\theta - \theta')\, g(\theta')\, p(\theta')\, \frac{d\theta'}{2\pi}.$$

Even when $\kappa$ depends only on $\theta - \theta'$, the extra factor $p(\theta')$ destroys shift invariance, so $T_p$ is no longer a pure convolution operator and its eigenfunctions need not be the global Fourier modes. As a result, a Fourier harmonic target can project onto multiple eigendirections of $T_p$, and "frequency" becomes distribution-dependent: the modes learned early are those aligned with the *top eigenfunctions of the density-weighted operator* rather than the lowest Fourier frequencies.

This dependence on $p$ is analyzed explicitly by Basri et al. (2020), who study the kernel regime on the circle/sphere and show that non-uniform densities can produce eigenfunctions with localized oscillatory structure and learning behavior that varies across the input space, with denser regions exhibiting effectively faster learning of finer-scale structure (Basri et al., 2020).

### 2.3.4 Spectral bias beyond the training set

Most early demonstrations of spectral bias are reported on the training set via Fourier spectra or projections onto empirical kernel eigenvectors. A natural question is whether a comparable phenomenon holds in function space more globally. Bowman and Montufar (2022) address this in the kernel regime by proving bounds that compare finite-width training trajectories to idealized kernel dynamics and conclude that networks inherit the bias of the

NTK integral operator over the input space, not merely at the training samples. In particular, they argue that eigenfunctions of the NTK integral operator are learned at rates corresponding to their eigenvalues, providing a mechanism for spectral bias that persists outside the training set (Bowman and Montufar, 2022).

### 2.3.5 Discussion and connection to feature learning

The operator-spectrum viewpoint provides a clean account of spectral bias in kernelized regimes, where learning rates are dictated by the spectrum of a fixed (or nearly fixed) operator. Outside the NTK regime, however, features can evolve substantially during optimization, and the relevant operator (e.g. the tangent kernel) may drift. From this perspective, spectral bias can be studied both as a *property of the initial linearized dynamics* and as a phenomenon that may interact with feature learning through time-dependent spectral structure. This interaction motivates empirical analyses that track mode-wise error decay alongside kernel evolution and provides a natural link between kernel-based and mean-field viewpoints.

## 2.4 Research gap

The preceding discussion shows that spectral bias is well understood in idealized kernel settings. In the infinite-width NTK regime, training is governed by a fixed operator whose spectrum determines the decay rates of the corresponding components of the residual (Jacot et al., 2020; Rahaman et al., 2019; Cao et al., 2020). On $S^1$ with the uniform measure, this operator becomes a convolution and is diagonalized by the Fourier basis. There is also substantial work on kernel convergence, finite-width corrections, and kernel evolution beyond the strict NTK regime (Golikov et al., 2022; Bordelon and Pehlevan, 2023; Vyas et al., 2022).

What is still missing is a perturbative analysis carried out within a single setting in which the continuum operator, its spectrum, and its eigenfunctions are all explicit. In particular, the existing theory does not yet provide a unified account of how the Fourier description on $S^1$ is altered first by replacing the continuum measure with finitely many uniform samples, and then by replacing the infinite-width kernel with the kernel induced by a finite-width network.

## 2.5 Research questions

This thesis studies spectral bias on $S^1$ by comparing the residual dynamics induced by the continuum infinite-width NTK operator with those induced by its finite-sample and finite-width counterparts.

**Main research question.** To what extent does the spectral decomposition of the continuum infinite-width NTK operator on $S^1$ persist under finite-sample and finite-width perturbations?

**Subquestions.**

(Q1) **Finite-sample perturbation of the continuum operator.** In the infinite-width regime, how does replacing the uniform measure on $S^1$ by $n$ i.i.d. sampled points affect the action of the continuum NTK operator on a fixed low-frequency Fourier subspace?

More precisely, how does the sampled operator differ from the continuum convolution operator on this subspace, and what are the consequences for the associated residual dynamics?

(Q2) **Finite-width perturbation in the continuum setting.** With the continuum measure on $S^1$ kept fixed, how does finite width perturb the limiting NTK operator and the associated residual dynamics?

More precisely, how does the kernel induced by a finite-width network differ from the continuum convolution operator, and how do these differences affect its spectrum, its Fourier eigenspaces, and the decay of residual components during training?

# Chapter 3

# Methodology

## 3.1 Mathematical setting

### 3.1.1 Domain and measure

We identify the unit circle $S^1$ with the angular domain $\Omega = [0, 2\pi)$, equipped with the uniform probability measure

$$d\mu(\varphi) = \frac{d\varphi}{2\pi}. \tag{3.1}$$

This parametrization makes the rotational symmetry of the problem explicit and provides the natural Fourier basis for $L^2(\mu)$.

### 3.1.2 Network model

We consider a one-hidden-layer ReLU network in NTK parameterization. For inputs $x \in S^1 \subset \mathbb{R}^2$,

$$f(x;\theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} a_i \sigma(w_i^\top x + b_i), \tag{3.2}$$

where $\theta = \{(a_i, w_i, b_i)\}_{i=1}^m$, $\sigma(t) = t_+ = \max\{t, 0\}$, and the parameters are initialized according to

$$a_i \sim \mathcal{N}(0,1), \qquad w_i \sim \mathcal{N}(0, I_2), \qquad b_i(0) = 0, \tag{3.3}$$

independently across $i = 1, \dots, m$. The bias parameters are initialized at zero but remain trainable throughout.

The analysis is carried out through the limiting neural tangent kernel in the infinite-width regime $m \to \infty$. This model is simple enough to admit an explicit kernel calculation on $S^1$, while still capturing the spectral structure relevant to the decay of Fourier components under training. The inclusion of trainable bias is part of this setup, since its contribution changes the Fourier spectrum of the limiting kernel. The derivation of the limiting kernel and the computation of its Fourier coefficients are deferred to Appendix B.

### 3.1.3 Residual dynamics and the kernel operator

We briefly recall the residual dynamics from Section 2.1.4, reformulated in the continuum gradient-flow setting used here. Let $y \in L^2(\mu)$ be the target function and let $f_t(\varphi) := f(\varphi;\theta(t))$ denote the network output at training time $t$. We consider the least-squares loss functional

$$\mathcal{L}(\theta) = \frac{1}{2}\|f(\cdot;\theta) - y\|^2_{L^2(\mu)} = \frac{1}{2}\int_0^{2\pi} \big(f(\varphi;\theta) - y(\varphi)\big)^2\, d\mu(\varphi), \tag{3.4}$$

whose functional derivative with respect to the output function is

$$\frac{\delta \mathcal{L}}{\delta f}(\varphi) = f(\varphi;\theta) - y(\varphi). \tag{3.5}$$

Accordingly, we define the residual

$$r_t(\varphi) = f_t(\varphi) - y(\varphi). \tag{3.6}$$

Under gradient flow in the regime where the limiting kernel is fixed, the residual evolves according to

$$\partial_t r_t(\varphi) = -\int_0^{2\pi} \Theta(\varphi,\psi)\, r_t(\psi)\, d\mu(\psi), \tag{3.7}$$

where $\Theta$ denotes the limiting neural tangent kernel. This leads to the integral operator $T : L^2(\mu) \to L^2(\mu)$ defined by

$$(Tf)(\varphi) = \int_0^{2\pi} \Theta(\varphi,\psi) f(\psi)\, d\mu(\psi), \tag{3.8}$$

so that (3.7) can be written as

$$\partial_t r_t = -T r_t. \tag{3.9}$$

Thus the decay of the residual is governed by the spectrum of $T$.

## 3.2 Continuum reference model

### 3.2.1 Translation invariance and Fourier diagonalization

The first object studied in the thesis is the limiting kernel operator on $S^1$. Because the input distribution is uniform and the kernel is rotation-invariant, $\Theta(\varphi,\psi)$ depends only on the angular difference. The operator (3.8) therefore takes the convolution form

$$(Tf)(\varphi) = \int_0^{2\pi} \Theta(\varphi - \psi) f(\psi)\, d\mu(\psi), \tag{3.10}$$

and is diagonalized by the real Fourier basis. Its eigenvalues determine the decay rates of the Fourier components of the residual.

This continuum Fourier decomposition is the starting point for the rest of the analysis. The finite-sample theory asks how much of this diagonal structure survives after replacing the measure $\mu$ by finitely many sampled points, while the later finite-width analysis asks how the behavior of the actual network departs from that of the limiting kernel. The explicit formula for the limiting kernel and the corresponding Fourier eigenvalues are stated in Chapter 4.6. Their derivation is deferred to Appendix B.

## 3.3 Finite-sample methodology

### 3.3.1 Finite-sample question

We now replace the uniform measure on $S^1$ by a finite sample

$$\varphi_1, \dots, \varphi_n \overset{\text{iid}}{\sim} \text{Unif}[0, 2\pi). \tag{3.11}$$

The main question is whether the diagonal Fourier decomposition of the continuum operator remains approximately valid after this replacement.

### 3.3.2 Restriction to a low-frequency Fourier subspace

Since the continuum operator is diagonalized by Fourier modes, the natural finite-dimensional approximation is obtained by truncating the real Fourier basis. We define

$$\phi_0(\varphi) = 1, \tag{3.12}$$

and for each $k \geq 1$,

$$\phi_{k,c}(\varphi) = \sqrt{2}\cos(k\varphi), \qquad \phi_{k,s}(\varphi) = \sqrt{2}\sin(k\varphi). \tag{3.13}$$

These functions form the standard real Fourier basis of $L^2(\mu)$.

For a fixed cutoff $K \geq 1$, we consider the truncated subspace

$$H_K = \text{span}\{\phi_0, \phi_{k,c}, \phi_{k,s} : 1 \leq k \leq K\}, \tag{3.14}$$

with dimension

$$d = 2K + 1. \tag{3.15}$$

This subspace isolates the low-frequency part of the spectrum and turns the comparison between the continuum and sampled operators into a finite-dimensional problem.

### 3.3.3 Sampled objects

To study the sampled operator on $H_K$, we introduce the feature matrix $\Phi \in \mathbb{R}^{n \times d}$,

$$\Phi_{i,p} = \phi_p(\varphi_i), \qquad 1 \leq i \leq n, \quad 1 \leq p \leq d, \tag{3.16}$$

and the normalized sampled kernel matrix

$$A_{ij} = \frac{1}{n}\Theta(\varphi_i - \varphi_j). \tag{3.17}$$

From these we form the Gram matrix

$$G = \frac{1}{n}\Phi^\top \Phi, \tag{3.18}$$

and the matrix representation of the sampled operator on the truncated Fourier basis,

$$H = \frac{1}{n}\Phi^{\top} A \Phi. \tag{3.19}$$

We also introduce the diagonal matrix

$$\Lambda^{(n)} = \operatorname{diag}(\lambda_p^{(n)}), \tag{3.20}$$

whose entries are the finite-sample corrected Fourier eigenvalues.

### 3.3.4 Quantities to be controlled

The finite-sample analysis is based on three comparisons: $G$ with the identity, $H$ with $\Lambda^{(n)}$, and the difference

$$A\Phi - \Phi\Lambda^{(n)}. \tag{3.21}$$

Controlling these quantities shows that, on the truncated Fourier subspace, the sampled operator remains close to the continuum Fourier decomposition.

# Chapter 4

# Results

## 4.1 Continuum kernel on $S^1$

We first state the explicit form of the limiting neural tangent kernel on the circle and the corresponding Fourier eigenvalues of the integral operator $T$ introduced in Chapter 3.3.4. The derivation is deferred to Appendix B.

**Proposition 4.1** (Limiting NTK on $S^1$)**.** *Let $\Theta$ be the limiting NTK associated with the model* (3.2)–(3.3)*. If $\delta \in [0,\pi]$ denotes the angular distance between two points on $S^1$, then $\Theta$ depends only on $\delta$ and is given by*

$$\Theta(\delta) = \frac{1}{2\pi}\Big(\sin\delta + 2(\pi-\delta)\cos\delta + (\pi-\delta)\Big). \tag{4.1}$$

*Consequently, the operator $T$ takes the convolution form*

$$(Tf)(\varphi) = \int_0^{2\pi} \Theta(\varphi-\psi) f(\psi)\, d\mu(\psi). \tag{4.2}$$

where $\Theta(\varphi-\psi)$ denotes the even $2\pi$-periodic extension of $\Theta(\delta)$.

Since $T$ is a convolution operator, it is diagonalized by the real Fourier basis introduced in (3.12)–(3.13).

**Theorem 4.2** (Fourier eigenvalues of the limiting NTK)**.** *Let $\lambda_k$ denote the eigenvalue of $T$ corresponding to the frequency $k$. Then*

$$\lambda_0 = \frac{1}{4} + \frac{3}{\pi^2}, \qquad \lambda_1 = \frac{1}{4} + \frac{1}{\pi^2}, \tag{4.3}$$

*and for $k \geq 2$,*

$$\lambda_k = \begin{cases} \dfrac{k^2+3}{\pi^2(k^2-1)^2}, & k \geq 2 \text{ even}, \\[2ex] \dfrac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd}. \end{cases} \tag{4.4}$$

*In particular, the decay rates of the Fourier components of the residual are determined by these eigenvalues.*

*Remark* 4.3 (Role of trainable bias). The trainable bias contributes an additional term to the limiting kernel and changes the parity structure of the spectrum. In particular, in the present model the odd Fourier modes have nonzero eigenvalues. By contrast, if the bias contribution is removed, the odd modes disappear from the fixed-kernel dynamics.

## 4.2 Finite-sample control on a truncated Fourier subspace

For each basis index $p$, let $k(p)$ denote the frequency of the corresponding basis function $\phi_p$, and write

$$\lambda_p := \lambda_{k(p)}.$$

We now turn to the sampled setting introduced in Chapter 3.3.4. Throughout, $K \geq 1$ is fixed, $d = 2K+1$, and the truncated Fourier subspace $H_K$ is given by (3.14). The matrices $\Phi$, $A$, $G$, and $H$ are defined in (3.16)–(3.19).

**Proposition 4.4** (Expectation of the restricted sampled operator)**.** *Let*

$$\lambda_p^{(n)} = \left(1 - \frac{1}{n}\right)\lambda_p + \frac{\Theta(0)}{n}, \tag{4.5}$$

*and let*

$$\Lambda^{(n)} = \operatorname{diag}(\lambda_p^{(n)}). \tag{4.6}$$

*Then*

$$\mathbb{E}[H] = \Lambda^{(n)}. \tag{4.7}$$

The next theorem gives explicit concentration bounds for the three sampled quantities that appear in the methodology chapter.

**Theorem 4.5** (Finite-sample concentration on the truncated Fourier subspace)**.** *Assume that*

$$\kappa := \sup_{\theta\in[0,2\pi)} |\Theta(\theta)| < \infty. \tag{4.8}$$

*Then, for every* $\varepsilon > 0$*,*

$$\mathbb{P}(\|G - I_d\|_{\max} \geq \varepsilon) \leq 2d^2 \exp\left(-\frac{n\varepsilon^2}{8}\right), \tag{4.9}$$

$$\mathbb{P}\left(\|H - \Lambda^{(n)}\|_{\max} \geq \varepsilon\right) \leq 2d^2 \exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right), \tag{4.10}$$

*and*

$$\mathbb{P}\left(\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \varepsilon\right) \leq 2nd \exp\left(-\frac{n\varepsilon^2}{4\kappa^2}\right). \tag{4.11}$$

**Corollary 4.6** (Simultaneous high-probability bounds)**.** *Let* $\delta \in (0,1)$*, and define*

$$\alpha_{n,d,\delta} := \sqrt{\frac{8}{n}\log\left(\frac{6d^2}{\delta}\right)}, \tag{4.12}$$

$$\beta_{n,d,\delta} := \kappa\sqrt{\frac{32}{n}\log\left(\frac{6d^2}{\delta}\right)}, \tag{4.13}$$

*and*

$$\gamma_{n,d,\delta} := \kappa\sqrt{\frac{4}{n}\log\left(\frac{6nd}{\delta}\right)}. \tag{4.14}$$

*Then, with probability at least* $1-\delta$*,*

$$\|G - I_d\|_{\max} \le \alpha_{n,d,\delta}, \qquad \|H - \Lambda^{(n)}\|_{\max} \le \beta_{n,d,\delta}, \qquad \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \le \gamma_{n,d,\delta}. \tag{4.15}$$

The previous theorem shows that the sampled operator remains close to the continuum Fourier decomposition on the truncated subspace $H_K$. To express this directly at the level of the induced matrix on coefficients, define

$$C := G^{-1}H, \tag{4.16}$$

whenever $G$ is invertible.

**Corollary 4.7** (Restricted sampled operator)**.** *Assume that* $d\,\alpha_{n,d,\delta} < 1$*. Then, on the event* (4.15)*, the matrix G is invertible and*

$$\|C - \Lambda^{(n)}\|_\infty \le \frac{d\,\beta_{n,d,\delta} + d\,\kappa\,\alpha_{n,d,\delta}}{1 - d\,\alpha_{n,d,\delta}}. \tag{4.17}$$

*In particular, the sampled operator restricted to $H_K$ remains close to the diagonal Fourier operator predicted by the continuum theory.*

Equations (4.9)–(4.17) show that, on any fixed truncated Fourier subspace, the sampled operator is a controlled perturbation of the continuum operator. Thus the ordering of low Fourier modes predicted by the continuum kernel persists with high probability after sampling.

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# Appendix A

# Mean-Field Viewpoint

## A.1 Detailed derivation of the mean-field representation

### A.1.1 Parameter space and empirical measure

Let $\Omega := \mathbb{R} \times \mathbb{R}^d$ denote the parameter space, and let $\mathcal{F}$ be its Borel $\sigma$-algebra. For each neuron $\alpha$, define the Dirac measure $\delta_{\theta_\alpha}$ on $(\Omega, \mathcal{F})$, i.e. if $A \in \mathcal{F}$ then,

$$\delta_{\theta_\alpha}(A) = \begin{cases} 1, & \theta_\alpha \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Define the empirical neuronal measure

$$\mu_m := \frac{1}{m} \sum_{\alpha=1}^{m} \delta_{\theta_\alpha}. \tag{A.1}$$

By construction, $\mu_m \geq 0$ and $\mu_m(\Omega) = 1$, hence $\mu_m \in \mathcal{P}(\Omega)$ where $\mathcal{P}(\Omega)$ is the space of Borel probability measures on $\Omega$.

**Integration against Dirac measures.** Let $\chi_A$ denote the indicator function of a measurable set $A \subset \Omega$. Then, for any $\theta_\alpha \in \Omega$,

$$\int_\Omega \chi_A(\theta)\, d\delta_{\theta_\alpha}(\theta) = \delta_{\theta_\alpha}(A) = \chi_A(\theta_\alpha).$$

More generally, if $s(\theta) = \sum_{k=1}^{r} c_k \chi_{A_k}(\theta)$ is a simple function, then

$$\int_\Omega s(\theta)\, d\delta_{\theta_\alpha}(\theta) = \sum_{k=1}^{r} c_k \chi_{A_k}(\theta_\alpha) = s(\theta_\alpha).$$

Any nonnegative measurable function $g : \Omega \to \mathbb{R}$ can be approximated by a sequence of simple functions, $\{s_n\}_{n\geq 0}$. If $s_n \uparrow g$ and each $s_n \geq 0$ then by the Monotone Convergence Theorem,

$$\int_\Omega g\, d\delta_{\theta_\alpha} = \int_\Omega \lim_{n\to\infty} s_n\, d\delta_{\theta_\alpha} = \lim_{n\to\infty} \int_\Omega s_n\, d\delta_{\theta_\alpha} = \lim_{n\to\infty} s_n(\theta_\alpha) = g(\theta_\alpha). \tag{A.2}$$

**Integral representation of the network.** Consider now the integral of $g$ with respect to the empirical measure $\mu_m$:

$$\int_\Omega g\,d\mu_m = \int_\Omega g\,d\left(\frac{1}{m}\sum_{\alpha=1}^{m}\delta_{\theta_\alpha}\right) = \frac{1}{m}\sum_{\alpha=1}^{m} g(\theta_\alpha).$$

Define

$$g(\theta_\alpha) = g(v_\alpha, w_\alpha) := v_\alpha\,\varphi(w_\alpha^\top x).$$

Then

$$\int_\Omega g\,d\mu_m = \frac{1}{m}\sum_{\alpha=1}^{m} v_\alpha\,\varphi(w_\alpha^\top x) = f_\theta(x), \tag{A.3}$$

recovering the finite-width network.

### A.1.2 Initialization and weak convergence

Assume that the parameters $\{\theta_\alpha(0)\}_{\alpha=1}^{m}$ are initialized i.i.d. according to a probability measure $\mu_0$ on $\Omega$. Define the empirical measure at initialization

$$\mu_{m,0} := \frac{1}{m}\sum_{\alpha=1}^{m}\delta_{\theta_\alpha(0)}.$$

Then $\mu_{m,0}$ converges weakly to $\mu_0$ almost surely as $m \to \infty$. Indeed, for any bounded continuous test function $\psi : \Omega \to \mathbb{R}$,

$$\int_\Omega \psi\,d\mu_{m,0} = \frac{1}{m}\sum_{\alpha=1}^{m}\psi(\theta_\alpha(0)) \xrightarrow[m\to\infty]{\text{a.s.}} \mathbb{E}_{\theta\sim\mu_0}[\psi(\theta)] = \int_\Omega \psi\,d\mu_0,$$

where the convergence follows from the strong law of large numbers applied to the i.i.d. random variables $\psi(\theta_\alpha(0))$. Since this holds for all bounded continuous $\psi$, we conclude that $\mu_{m,0} \to \mu_0$ (Varadarajan, 1958). As a consequence, the network output at initialization converges pointwise to the deterministic limit

$$f_{\mu_0}(x) := \int_\Omega v\,\varphi(w^\top x)\,d\mu_0(v,w).$$

For instance, under i.i.d. Gaussian initialization of the parameters, i.e. $v_0 \sim \mathcal{N}(0, I_m)\ W_0 \sim \mathcal{N}(0, I_m \otimes I_d)$, the limiting initial measure $\mu_0 = \mathcal{N}(0, I_{d+1})$ is a Gaussian measure on a $d+1$-dimensional space.

### A.1.3 Training dynamics and evolution of the empirical measure

Let $X = (x_1, \ldots, x_n) \in \mathbb{R}^{d\times n}$ and $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$ denote the training set, and consider the squared loss

$$L(v,W) = \frac{1}{2}\sum_{i=1}^{n}\big(f_{v,W}(x_i) - y_i\big)^2 = \frac{1}{2}\|f_{v,W}(X) - y\|_2^2,$$

where $f_{v,W}(X) := (f_{v,W}(x_i))_{i=1}^n$.

We study continuous-time gradient flow for a finite-width network with $m$ neurons and learning rate $\eta > 0$,

$$\dot{v}_{t,\alpha} = -\eta \nabla_{v_\alpha} L(v_t, W_t), \qquad \dot{w}_{t,\alpha} = -\eta \nabla_{w_\alpha} L(v_t, W_t), \qquad \alpha \in \{1, \ldots, m\}.$$

A direct computation yields, for each $\alpha$,

$$\dot{v}_{t,\alpha} = \frac{\eta}{m} \varphi(X^\top w_{t,\alpha})^\top \big(y - f_{v_t,W_t}(X)\big), \tag{A.4}$$

$$\dot{w}_{t,\alpha} = \frac{\eta}{m} X \Big(\varphi'(X^\top w_{t,\alpha}) \odot \big(y - f_{v_t,W_t}(X)\big)\Big) v_{t,\alpha}, \tag{A.5}$$

where $\odot$ denotes elementwise (Hadamard) multiplication.

As seen in (A.4)–(A.5), evolution of each neuron depends on the parameters only through the current empirical measure $\mu_{m,t}$ via $f_{v_t,W_t}(X)$. Thus, the particle system (A.4)–(A.5) induces an evolution of the empirical measure in parameter space (Mei et al., 2018; Rotskoff and Vanden-Eijnden, 2022).

Hence, the particle system (A.4)–(A.5) can be viewed as an interacting particle system driven by a measure-dependent velocity field. For any $\theta = (v, w) \in \Omega$ and any $\mu \in \mathcal{P}(\Omega)$, define

$$b((v,w);\mu) := \begin{pmatrix} \varphi(X^\top w)^\top \big(y - f_\mu(X)\big) \\ X\Big(\varphi'(X^\top w) \odot \big(y - f_\mu(X)\big)\Big) v \end{pmatrix},$$

where

$$f_\mu(x) := \int_\Omega v \varphi(w^\top x)\, d\mu(v,w), \qquad f_\mu(X) := \big(f_\mu(x_i)\big)_{i=1}^n.$$

Then the gradient-flow dynamics can be written compactly as

$$\dot{\theta}_{t,\alpha} = \frac{\eta}{m} b(\theta_{t,\alpha}; \mu_{m,t}), \qquad \alpha = 1, \ldots, m.$$

In particular, each particle interacts with the rest of the system only through the current empirical measure $\mu_{m,t}$ (via $f_{\mu_{m,t}}(X)$).

### A.1.4 Continuity equation and push-forward formulation

Equivalently, $\mu_{m,t}$ solves the continuity equation

$$\partial_t \mu_{m,t} + \nabla_\theta \cdot \Big(\mu_{m,t} \frac{\eta}{m} b(\cdot; \mu_{m,t})\Big) = 0, \tag{A.6}$$

in the sense of distributions, i.e. for every test function $\psi \in C_c^\infty(\Omega)$,

$$\frac{d}{dt} \int_\Omega \psi(\theta)\, d\mu_{m,t}(\theta) = \frac{\eta}{m} \int_\Omega \nabla_\theta \psi(\theta) \cdot b(\theta; \mu_{m,t})\, d\mu_{m,t}(\theta).$$

Choosing the mean-field scaling $\eta = m$ yields a nontrivial $O(1)$ evolution in time and, as $m \to \infty$, one expects $\mu_{m,t} \to \mu_t$, where the limit $\mu_t$ satisfies

$$\partial_t \mu_t + \nabla_\theta \cdot \big(\mu_t\, b(\cdot; \mu_t)\big) = 0, \qquad \mu_0 = \mathcal{N}(0, I_{d+1}). \tag{A.7}$$

**Push-forward formulation.** Recall that if $g : \mathcal{X} \to \mathcal{Z}$ is measurable and $\mu \in \mathcal{P}(\mathcal{X})$, then the *push-forward* of $\mu$ by $g$ is the measure $g_*\mu \in \mathcal{P}(\mathcal{Z})$ defined by

$$(g_*\mu)(A) := \mu\big(g^{-1}(A)\big), \qquad A \subset \mathcal{Z} \text{ measurable.}$$

Let $\Phi_t : \Omega \to \Omega$ denote the flow map associated with the velocity field $\frac{\eta}{m} b(\cdot; \mu_{m,t})$. Then the empirical measure is transported by this flow:

$$\mu_{m,t} = (\Phi_t)_* \mu_{m,0}.$$

In particular, for any measurable observable $h : \Omega \to \mathbb{R}$,

$$\int_\Omega h(\theta)\, d\mu_{m,t}(\theta) = \int_\Omega h(\Phi_t(\theta))\, d\mu_{m,0}(\theta),$$

so evaluating the network at time $t$ is obtained by taking $h_x(\theta) := v\varphi(w^\top x)$ and writing

$$f_{\mu_{m,t}}(x) = \int_\Omega v\varphi(w^\top x)\, d\mu_{m,t}(v,w) = \int_\Omega v\varphi(w^\top x)\, d\big((\Phi_t)_*\mu_{m,0}\big)(v,w).$$

# Appendix B

# Derivation of the limiting kernel on $S^1$

This appendix derives the explicit form of the limiting neural tangent kernel on the circle for the model introduced in Chapter 3.3.4, and computes the corresponding Fourier eigenvalues of the integral operator $T$. These derivations justify Proposition 4.1 and Theorem 4.2.

## B.1 Decomposition of the empirical neural tangent kernel

Recall the network

$$f(x;\theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} a_i \sigma(w_i^\top x + b_i),$$

where $x \in S^1 \subset \mathbb{R}^2$, $\sigma(t) = t_+$, and $\theta = \{(a_i, w_i, b_i)\}_{i=1}^m$. For $x, y \in S^1$, the empirical neural tangent kernel is

$$\Theta_m(x,y) = \big\langle \nabla_\theta f(x;\theta), \nabla_\theta f(y;\theta) \big\rangle. \tag{B.1}$$

For each hidden unit $i$, define

$$u_i := w_i^\top x + b_i, \qquad v_i := w_i^\top y + b_i.$$

Then

$$\frac{\partial f(x;\theta)}{\partial a_i} = \frac{1}{\sqrt{m}} \sigma(u_i), \qquad \frac{\partial f(x;\theta)}{\partial w_i} = \frac{1}{\sqrt{m}} a_i \sigma'(u_i) x, \qquad \frac{\partial f(x;\theta)}{\partial b_i} = \frac{1}{\sqrt{m}} a_i \sigma'(u_i).$$

Substituting into (B.1) yields

$$\Theta_m(x,y) = \Theta_m^{(a)}(x,y) + \Theta_m^{(w)}(x,y) + \Theta_m^{(b)}(x,y), \tag{B.2}$$

where

$$\Theta_m^{(a)}(x,y) = \frac{1}{m}\sum_{i=1}^{m}\sigma(u_i)\sigma(v_i),$$
$$\Theta_m^{(w)}(x,y) = \frac{1}{m}\sum_{i=1}^{m}a_i^2\sigma'(u_i)\sigma'(v_i)\,x^\top y,$$
$$\Theta_m^{(b)}(x,y) = \frac{1}{m}\sum_{i=1}^{m}a_i^2\sigma'(u_i)\sigma'(v_i).$$

## B.2 Infinite-width limit

At initialization, the parameters satisfy

$$a_i \sim \mathcal{N}(0,1), \qquad w_i \sim \mathcal{N}(0,I_2), \qquad b_i(0) = 0,$$

independently across $i$. Since the hidden units are independent and identically distributed, the law of large numbers implies that $\Theta_m(x,y)$ converges almost surely, as $m\to\infty$, to a deterministic kernel $\Theta(x,y)$.

Because $b_i(0) = 0$, the limiting kernel is

$$\Theta(x,y) = \mathbb{E}\big[\sigma(u)\sigma(v)\big] + \big(x^\top y + 1\big)\mathbb{E}\big[\sigma'(u)\sigma'(v)\big], \tag{B.3}$$

where

$$u = w^\top x, \qquad v = w^\top y, \qquad w \sim \mathcal{N}(0,I_2).$$

We write

$$K_0(x,y) := \mathbb{E}\big[\sigma(u)\sigma(v)\big], \qquad K_1(x,y) := \mathbb{E}\big[\sigma'(u)\sigma'(v)\big],$$

so that

$$\Theta(x,y) = K_0(x,y) + \big(x^\top y + 1\big)K_1(x,y).$$

## B.3 Restriction to the circle

Write

$$x(\varphi) = (\cos\varphi, \sin\varphi), \qquad x(\psi) = (\cos\psi, \sin\psi),$$

and let $\delta \in [0,\pi]$ denote the angle between $x(\varphi)$ and $x(\psi)$, so that

$$x(\varphi)^\top x(\psi) = \cos\delta.$$

By rotational invariance of the Gaussian measure, the kernel depends only on $\delta$. We therefore write $\Theta(\delta)$, $K_0(\delta)$, and $K_1(\delta)$.

To compute these quantities, we rotate coordinates so that

$$x = (1,0), \qquad y = (\cos\delta, \sin\delta).$$

If $w = (Z_1, Z_2)$ with $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, then

$$u = Z_1, \qquad v = Z_1\cos\delta + Z_2\sin\delta.$$

**Lemma B.1.** *For* $\delta \in [0,\pi]$,

$$K_1(\delta) = \mathbb{P}(u > 0, v > 0) = \frac{\pi - \delta}{2\pi}. \tag{B.4}$$

*Proof.* Since $(u,v)$ is a centered bivariate Gaussian with correlation $\cos\delta$, the probability that both coordinates are positive is the standard Gaussian quadrant probability, which equals $(\pi-\delta)/(2\pi)$. ☐

**Lemma B.2.** *For* $\delta \in [0,\pi]$,

$$K_0(\delta) = \mathbb{E}[\sigma(u)\sigma(v)] = \frac{1}{2\pi}\Big(\sin\delta + (\pi-\delta)\cos\delta\Big). \tag{B.5}$$

*Proof.* This is the standard arc-cosine covariance formula for ReLU features in two dimensions. ☐

*Proof of Proposition 4.1.* Using (B.3), (B.4), and (B.5), we obtain

$$\begin{aligned}\Theta(\delta) &= \frac{1}{2\pi}\Big(\sin\delta + (\pi-\delta)\cos\delta\Big) + \frac{1+\cos\delta}{2\pi}(\pi-\delta)\\ &= \frac{1}{2\pi}\Big(\sin\delta + 2(\pi-\delta)\cos\delta + (\pi-\delta)\Big).\end{aligned} \tag{B.6}$$

This is exactly (4.1). The convolution form of the operator follows from the fact that $\Theta$ depends only on the angular difference. ☐

## B.4 Fourier coefficients of the limiting kernel

Because the operator $T$ is a convolution operator, it is diagonalized by the real Fourier basis. If $\lambda_k$ denotes the eigenvalue associated with frequency $k$, then

$$\lambda_k = \frac{1}{\pi}\int_0^\pi \Theta(\delta)\cos(k\delta)\,d\delta, \qquad k \geq 0. \tag{B.7}$$

Substituting (B.6) gives

$$\lambda_k = \frac{1}{2\pi^2}\int_0^\pi \Big(\sin\delta + 2(\pi-\delta)\cos\delta + (\pi-\delta)\Big)\cos(k\delta)\,d\delta.$$

We first record a basic integration lemma.

**Lemma B.3.** *For every integer* $m \geq 1$,

$$I_m := \int_0^\pi (\pi-\delta)\cos(m\delta)\,d\delta = \frac{1-(-1)^m}{m^2}. \tag{B.8}$$

*Proof.* Integrating by parts gives

$$I_m = \left[(\pi-\delta)\frac{\sin(m\delta)}{m}\right]_0^\pi + \frac{1}{m}\int_0^\pi \sin(m\delta)\,d\delta.$$

The boundary term vanishes, and

$$\frac{1}{m}\int_0^\pi \sin(m\delta)\,d\delta = \frac{1}{m}\left[-\frac{\cos(m\delta)}{m}\right]_0^\pi = \frac{1-(-1)^m}{m^2}.$$

□

*Proof of Theorem 4.2.* For $k \geq 2$, write

$$\lambda_k = \frac{1}{2\pi^2}(A_k + B_k + C_k),$$

where

$$\begin{aligned} A_k &:= \int_0^\pi \sin\delta\,\cos(k\delta)\,d\delta, \\ B_k &:= \int_0^\pi 2(\pi-\delta)\cos\delta\cos(k\delta)\,d\delta, \\ C_k &:= \int_0^\pi (\pi-\delta)\cos(k\delta)\,d\delta. \end{aligned}$$

Using

$$\sin\delta\cos(k\delta) = \frac{1}{2}\big(\sin((k+1)\delta) - \sin((k-1)\delta)\big),$$

we obtain

$$A_k = \frac{1}{2}\left(\frac{1-(-1)^{k+1}}{k+1} - \frac{1-(-1)^{k-1}}{k-1}\right).$$

Using

$$2\cos\delta\cos(k\delta) = \cos((k+1)\delta) + \cos((k-1)\delta),$$

together with Lemma B.3, we obtain

$$B_k = \frac{1-(-1)^{k+1}}{(k+1)^2} + \frac{1-(-1)^{k-1}}{(k-1)^2}, \qquad C_k = \frac{1-(-1)^k}{k^2}.$$

If $k \geq 3$ is odd, then $k \pm 1$ are even, so $A_k = B_k = 0$, while $C_k = 2/k^2$. Hence

$$\lambda_k = \frac{1}{\pi^2 k^2}, \qquad k \geq 3 \text{ odd}. \tag{B.9}$$

If $k \geq 2$ is even, then $k \pm 1$ are odd, so $C_k = 0$, and

$$A_k = -\frac{2}{k^2-1}, \qquad B_k = \frac{4(k^2+1)}{(k^2-1)^2}.$$

Therefore
$$A_k + B_k = \frac{2(k^2+3)}{(k^2-1)^2},$$
and so
$$\lambda_k = \frac{k^2+3}{\pi^2(k^2-1)^2}, \qquad k \geq 2 \text{ even.} \tag{B.10}$$

It remains to compute $k = 0$ and $k = 1$ directly. For $k = 0$,
$$\lambda_0 = \frac{1}{2\pi^2}\left(\int_0^\pi \sin\delta\, d\delta + 2\int_0^\pi (\pi-\delta)\cos\delta\, d\delta + \int_0^\pi (\pi-\delta)\, d\delta\right) = \frac{1}{4} + \frac{3}{\pi^2}.$$
For $k = 1$,
$$\lambda_1 = \frac{1}{2\pi^2}\left(\int_0^\pi \sin\delta\cos\delta\, d\delta + 2\int_0^\pi (\pi-\delta)\cos^2\delta\, d\delta + \int_0^\pi (\pi-\delta)\cos\delta\, d\delta\right) = \frac{1}{4} + \frac{1}{\pi^2}.$$
This proves (4.3) and (4.4). □

## B.5 Effect of the bias contribution

We conclude by isolating the contribution of the trainable bias. From (B.2), the bias term contributes
$$\Theta^{(b)}(x,y) = \mathbb{E}\left[\sigma'(u)\sigma'(v)\right].$$
On the circle, this gives the term $(\pi-\delta)/(2\pi)$. In the Fourier coefficients, this is precisely the term responsible for the nonzero odd eigenvalues. If the trainable bias is removed, then $\Theta^{(b)}$ disappears and the odd frequencies no longer contribute to the fixed-kernel dynamics, which proves Remark 4.3.

# Appendix C

# Proofs of the finite-sample estimates

This appendix proves the finite-sample results from Section 4.2. We work with the truncated Fourier subspace $H_K$, the feature matrix $\Phi$, the sampled kernel matrix $A$, the Gram matrix $G$, and the matrix $H$ defined in (3.16)–(3.19).

## C.1 Preliminaries

For each basis function in (3.12)–(3.13),

$$|\phi_p(\varphi)| \leq \sqrt{2} \qquad \text{for all } \varphi \in [0, 2\pi), \quad 1 \leq p \leq d. \tag{C.1}$$

We also write

$$\kappa := \sup_{\theta \in [0,2\pi)} |\Theta(\theta)| < \infty. \tag{C.2}$$

For each basis index $p$, let $\lambda_p$ denote the eigenvalue of the operator $T$ associated with $\phi_p$, so that

$$T\phi_p = \lambda_p \phi_p.$$

Define the finite-sample corrected eigenvalues by

$$\lambda_p^{(n)} = \left(1 - \frac{1}{n}\right)\lambda_p + \frac{\Theta(0)}{n}, \tag{C.3}$$

and let

$$\Lambda^{(n)} = \operatorname{diag}(\lambda_p^{(n)}).$$

## C.2 Expectation of $H$

*Proof of Proposition 4.4.* By definition,

$$H_{pq} = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \phi_p(\varphi_i)\Theta(\varphi_i - \varphi_j)\phi_q(\varphi_j). \tag{C.4}$$

Split the sum into the cases $i \neq j$ and $i = j$:

$$H_{pq} = \frac{1}{n^2}\sum_{i\neq j}\phi_p(\varphi_i)\Theta(\varphi_i-\varphi_j)\phi_q(\varphi_j) + \frac{1}{n^2}\sum_{i=1}^{n}\phi_p(\varphi_i)\Theta(0)\phi_q(\varphi_i).$$

For the diagonal terms,

$$\mathbb{E}\left[\frac{1}{n^2}\sum_{i=1}^{n}\phi_p(\varphi_i)\Theta(0)\phi_q(\varphi_i)\right] = \frac{\Theta(0)}{n}\delta_{pq}.$$

For the off-diagonal terms, independence of $\varphi_i$ and $\varphi_j$ gives

$$\begin{aligned}
\mathbb{E}[\phi_p(\varphi_i)\Theta(\varphi_i-\varphi_j)\phi_q(\varphi_j)] &= \int_0^{2\pi}\phi_p(\varphi)\left(\int_0^{2\pi}\Theta(\varphi-\psi)\phi_q(\psi)\,d\mu(\psi)\right)d\mu(\varphi)\\
&= \int_0^{2\pi}\phi_p(\varphi)(T\phi_q)(\varphi)\,d\mu(\varphi) = \lambda_q\delta_{pq}.
\end{aligned}$$

Since there are $n(n-1)$ off-diagonal pairs, we conclude that

$$\mathbb{E}[H_{pq}] = \left(1-\frac{1}{n}\right)\lambda_q\delta_{pq} + \frac{\Theta(0)}{n}\delta_{pq} = \lambda_q^{(n)}\delta_{pq}.$$

Therefore $\mathbb{E}[H] = \Lambda^{(n)}$. □

## C.3 Concentration of the Gram matrix

*Proof of* (4.9). For fixed $p, q$,

$$G_{pq} = \frac{1}{n}\sum_{i=1}^{n} Z_i, \qquad Z_i := \phi_p(\varphi_i)\phi_q(\varphi_i).$$

The variables $Z_i$ are independent and identically distributed, with

$$\mathbb{E}[Z_i] = \delta_{pq}$$

by orthonormality of the Fourier basis. By (C.1),

$$|Z_i| \leq 2,$$

so each $Z_i$ takes values in an interval of length 4. Hoeffding's inequality gives

$$\mathbb{P}(|G_{pq}-\delta_{pq}| \geq \varepsilon) \leq 2\exp\left(-\frac{n\varepsilon^2}{8}\right).$$

Taking a union bound over all $d^2$ entries yields

$$\mathbb{P}(\|G-I_d\|_{\max} \geq \varepsilon) \leq 2d^2\exp\left(-\frac{n\varepsilon^2}{8}\right).$$

□

## C.4 Concentration of $H$

*Proof of* (4.10). Fix $p,q$, and write $H_{pq} = F(\varphi_1,\dots,\varphi_n)$, where

$$F(\varphi_1,\dots,\varphi_n) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\phi_p(\varphi_i)\Theta(\varphi_i-\varphi_j)\phi_q(\varphi_j).$$

We apply McDiarmid's inequality. If only one coordinate, say $\varphi_s$, is changed, then only the terms with $i=s$ or $j=s$ can change. There are at most $2n$ such terms. By (C.1) and (C.2), each summand has absolute value at most

$$|\phi_p(\cdot)\Theta(\cdot)\phi_q(\cdot)| \le \sqrt{2}\,\kappa\sqrt{2} = 2\kappa,$$

so one summand changes by at most $4\kappa$. Therefore

$$|F(\varphi_1,\dots,\varphi_s,\dots,\varphi_n) - F(\varphi_1,\dots,\varphi_s',\dots,\varphi_n)| \le \frac{8\kappa}{n}.$$

McDiarmid's inequality then gives

$$\mathbb{P}(|H_{pq}-\mathbb{E}[H_{pq}]| \ge \varepsilon) \le 2\exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right).$$

A union bound over all $d^2$ entries yields

$$\mathbb{P}\Big(\|H-\Lambda^{(n)}\|_{\max} \ge \varepsilon\Big) \le 2d^2\exp\left(-\frac{n\varepsilon^2}{32\kappa^2}\right).$$

□

## C.5 Control of the difference $A\Phi - \Phi\Lambda^{(n)}$

*Proof of* (4.11). Fix $i,p$. By definition,

$$(A\Phi)_{i,p} = \frac{1}{n}\sum_{j=1}^{n}\Theta(\varphi_i-\varphi_j)\phi_p(\varphi_j).$$

Split off the diagonal term $j=i$:

$$(A\Phi)_{i,p} = \frac{\Theta(0)}{n}\phi_p(\varphi_i) + \frac{1}{n}\sum_{j\ne i}\Theta(\varphi_i-\varphi_j)\phi_p(\varphi_j).$$

Condition on $\varphi_i$. For $j\ne i$, define

$$X_j := \Theta(\varphi_i-\varphi_j)\phi_p(\varphi_j).$$

Then $X_j$ are conditionally independent and identically distributed, and

$$|X_j| \le \kappa\sqrt{2}$$

by (C.1) and (C.2). Moreover,

$$\mathbb{E}[X_j \mid \varphi_i] = \int_0^{2\pi} \Theta(\varphi_i - \psi)\phi_p(\psi)\, d\mu(\psi) = (T\phi_p)(\varphi_i) = \lambda_p \phi_p(\varphi_i).$$

Therefore

$$\mathbb{E}[(A\Phi)_{i,p} \mid \varphi_i] = \lambda_p^{(n)} \phi_p(\varphi_i).$$

Apply Hoeffding's inequality to the average of the $n-1$ variables $X_j$. Since they lie in an interval of length $2\kappa\sqrt{2}$, we obtain

$$\mathbb{P}\Big( \Big| (A\Phi)_{i,p} - \lambda_p^{(n)} \phi_p(\varphi_i) \Big| \geq \varepsilon \;\Big|\; \varphi_i \Big) \leq 2\exp\left( -\frac{n\varepsilon^2}{4\kappa^2} \right),$$

where we used $n \geq 2$ to simplify the constant. Removing the conditioning and taking a union bound over the $nd$ entries yields

$$\mathbb{P}\Big( \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \varepsilon \Big) \leq 2nd \exp\left( -\frac{n\varepsilon^2}{4\kappa^2} \right).$$

□

## C.6 Proof of the simultaneous high-probability bound

*Proof of Theorem 4.6.* Set

$$\alpha_{n,d,\delta} = \sqrt{\frac{8}{n}\log\left(\frac{6d^2}{\delta}\right)}, \qquad \beta_{n,d,\delta} = \kappa\sqrt{\frac{32}{n}\log\left(\frac{6d^2}{\delta}\right)}, \qquad \gamma_{n,d,\delta} = \kappa\sqrt{\frac{4}{n}\log\left(\frac{6nd}{\delta}\right)}.$$

By the bounds proved in Sections C.3–C.5,

$$\mathbb{P}\big( \|G - I_d\|_{\max} \geq \alpha_{n,d,\delta} \big) \leq \frac{\delta}{3},$$

$$\mathbb{P}\Big( \|H - \Lambda^{(n)}\|_{\max} \geq \beta_{n,d,\delta} \Big) \leq \frac{\delta}{3},$$

and

$$\mathbb{P}\Big( \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \gamma_{n,d,\delta} \Big) \leq \frac{\delta}{3}.$$

A union bound gives the result. □

## C.7 Proof of the corollary on the restricted sampled operator

*Proof of Corollary 4.7.* Recall that

$$C := G^{-1}H.$$

On the event of Theorem 4.6,

$$\|G - I_d\|_\infty \le d\,\alpha_{n,d,\delta}, \qquad \|H - \Lambda^{(n)}\|_\infty \le d\,\beta_{n,d,\delta}, \tag{C.5}$$

since $\|M\|_\infty \le d\|M\|_{\max}$ for every $d \times d$ matrix $M$.

If $d\,\alpha_{n,d,\delta} < 1$, then $G$ is invertible and

$$\|G^{-1}\|_\infty \le \frac{1}{1 - d\,\alpha_{n,d,\delta}} \tag{C.6}$$

by the Neumann-series estimate.

Using

$$C - \Lambda^{(n)} = G^{-1}\big((H - \Lambda^{(n)}) - (G - I_d)\Lambda^{(n)}\big), \tag{C.7}$$

we obtain

$$\|C - \Lambda^{(n)}\|_\infty \le \|G^{-1}\|_\infty \left(\|H - \Lambda^{(n)}\|_\infty + \|G - I_d\|_\infty \|\Lambda^{(n)}\|_\infty\right).$$

Since $|\lambda_p| \le \kappa$ and $|\Theta(0)| \le \kappa$, the definition of $\lambda_p^{(n)}$ implies

$$\|\Lambda^{(n)}\|_\infty \le \kappa.$$

Substituting (C.5) and (C.6) gives

$$\|C - \Lambda^{(n)}\|_\infty \le \frac{d\,\beta_{n,d,\delta} + d\,\kappa\,\alpha_{n,d,\delta}}{1 - d\,\alpha_{n,d,\delta}},$$

which is exactly (4.17). □